Software Requirements Specification (SRS)
ResearchRAG AI – Intelligent Academic Research Assistant

1. Project Name & Background

- **Project Name:** ResearchRAG AI
- **Background:** Researchers face difficulty finding relevant papers, classifying topics, and extracting insights efficiently. Traditional keyword search lacks semantic understanding and citation grounding.
- **Problem Statement:** Current tools lack an integrated solution for combining semantic search, automatic classification, and citation-grounded question answering in one system.

2. Objectives:

a. **Paper Classification:** Title + abstract → category prediction (ML models)
b. **Research Q&A (RAG):** Retrieve papers → citation-based answers (Gemini API)
c. **Semantic Search:** Embedding-based similarity search
d. **Fallback Mode:** Local predictions + retrieved context if API unavailable
e. **Interface:** Streamlit web app

3. Scope & Solution

Aspect	Details
Scope	arXiv dataset, paper classification, semantic retrieval, Q&A, evaluation, visualization
ML Models	TF-IDF + Logistic Regression TF-IDF + MLP Classifier Metrics: Accuracy, Precision, Recall, F1-score
RAG System	Embeddings: sentence-transformers/all-MiniLM-L6-v2 Vector DB: FAISS Answer Generation: Google Gemini API Fallback: trained local models + retrieval-grounded evidence
Tech Stack	Streamlit, Python, Scikit-learn, FAISS, Google Gemini API, sentence-transformers

4. Comparison with Existing Projects

- **PaperQA2:** Research paper Q&A with citations (https://github.com/whitead/paper-qa)
- **Elicit:** Literature search & summarization (https://elicit.com)

Edge of ResearchRAG AI:

a. Combines ML classification + RAG
b. Citation-grounded answers
c. Offline fallback
d. Fully deployable academic assistant

5. Dataset

a. **Kaggle:** https://www.kaggle.com/datasets/Cornell-University/arxiv

6. Deliverables & Resources

Deliverables:

a. GitHub repo: code, notebooks, reports, documentation
b. Streamlit App: classification + Q&A
c. Saved local model artifacts (.pkl) used by the app
d. EDA & evaluation reports
e. Final report + slides

Resources:

f. GitHub: https://github.com/Rameesha8/ResearchRAG-AI
g. Deployment: Streamlit Community Cloud
h. Gemini API Docs: https://ai.google.dev/gemini-api/docs/quickstart